

SAILESHKUMAR A

AI Engineer with hands-on experience in Machine Learning, Deep Learning, and Python-based model development

+91 9629320388 | sailesh4246V@gmail.com | [linkedin.com/in/sailesh-kumar-a571862b2](https://www.linkedin.com/in/sailesh-kumar-a571862b2) | github.com/SAILESHKUMAR33 | Chennai, India

PROFESSIONAL SUMMARY

- B.Tech graduate in Artificial Intelligence and Data Science with practical experience across machine learning, deep learning, computer vision, and applied generative AI / LLM projects, built primarily in Python.
- Have trained and evaluated a range of models — CNNs (VGG16/VGG19), MLPs, YOLOv8 object detectors, and classical models like RandomForest — for classification, detection, and forecasting problems.
- Comfortable with the full ML lifecycle: data cleaning and feature engineering, model training and evaluation, and deploying the result as something people can actually use (Streamlit apps, dashboards, APIs).
- Have also built LLM-based systems using the Gemini API — prompt engineering, structured JSON generation, embeddings, and retrieval-augmented generation (RAG) for semantic search.
- Currently an AI Intern applying these skills in a live product environment; looking to move into a full-time AI/ML Engineer role.

EXPERIENCE

AI Intern | Codework.ai

3 Months (Current) | On-site

- Building and integrating AI/ML-driven features into product workflows, working closely with engineers on model integration and testing.
- Writing Python code to prepare data, prototype ML components, and evaluate model outputs against product requirements.
- Collaborating with senior engineers on model debugging, code reviews, and iterative experimentation in an agile environment.
- Using Git for version control and contributing to shared repositories following team coding standards.
- Documenting experiments, model performance, and technical decisions for internal knowledge sharing.
- Applied real-time computer vision skills from this role to independently design and build a YOLOv8-based fight/threat-detection prototype (see Fight-detection project below), including model benchmarking, DeepSORT-based tracking, and a deployed Streamlit monitoring dashboard.

Data Science Intern | Young's Mind Technology Solutions Pvt Ltd

1 Month | On-site

- Developed and evaluated machine learning models in Python for predictive analytics and data-driven decision-making.
- Performed data cleaning, preprocessing, and feature engineering to improve model performance.
- Conducted exploratory data analysis (EDA) to identify trends, patterns, and anomalies in datasets.
- Implemented data validation techniques to ensure data quality, consistency, and accuracy for model inputs.
- Automated repetitive data workflows by developing reusable Python scripts, improving processing efficiency.
- Built interactive dashboards and visualizations (Power BI, Matplotlib) to communicate model results and key findings.
- Used Git and GitHub for version control and collaborative development with cross-functional teams.

PROJECTS

Fight-detection — Real-Time Threat Detection System — github.com/SAILESHKUMAR33/Fight-detection

- Built a real-time video-surveillance pipeline combining a pretrained YOLOv8 violence-detection model with a DeepSORT person tracker and rule-based loitering logic (duration + drift), deployed via a modular architecture (camera, detection, tracking, behavior, alerts, dashboard).
- Solved the cold-start problem of not having labeled fight-video data by evaluating a pretrained YOLOv8 detector against a custom GRU classifier trained on MediaPipe pose keypoints, letting the team ship a working detector before custom training data was ready.
- Built a model-benchmarking workflow (benchmark_models.py, ROC/AUC curve generation) to compare detection approaches objectively and justify the production model choice — [add your measured accuracy/precision/recall/AUC figures here].
- Deployed the system as a Streamlit web dashboard with a live-alerts view and an upload-and-analyze mode, backed by a SQLite database for persisting and querying threat alerts.
- Designed the codebase as a standalone, swappable variant of a larger threat-alert-ai system, so the fight-detection model, tracker, and dashboard can each be replaced independently — documented setup and Colab-based GPU training in project READMEs for reproducibility.

Autism Detection using VGG16/VGG19 (EEG Data) — github.com/SAILESHKUMAR33/projectphase2

- Built a deep learning pipeline using VGG16 and VGG19 architectures to classify EEG electrode data for early autism detection.
- Performed data preprocessing, model training, and evaluation; benchmarked both architectures on accuracy, precision, recall, and F1-score.
- Tuned hyperparameters and model configuration to optimize performance for reliable, real-world classification.

Handwritten Digit Classifier (MLP)

- Designed and trained a Multi-Layer Perceptron neural network from scratch to classify handwritten digits.

- Achieved high classification accuracy through systematic training, validation, and hyperparameter tuning.

Comment Classification System (OpenCV / Python)

- Built a computer vision-based classification pipeline using OpenCV to categorize comment data for sentiment analysis.
- Designed preprocessing modules that improved model accuracy and reduced noise in input data.

Semantic Analysis Engine (NLP)

- Developed a Python-based NLP tool to perform semantic analysis and extract structured insights from unstructured text.
- Applied text-processing techniques to improve the accuracy of automated interpretation.

Vending Machine Sales Analytics (Streamlit, scikit-learn) — github.com/SAILESHKUMAR33/vending-machine-project

- Built a multi-page Streamlit analytics app that turns raw vending-machine sales exports into category/product summaries, a demand forecast, a refill plan, and dynamic pricing recommendations.
- Trained a RandomForest regression model (scikit-learn) on weekly lag features — lag-1, lag-2, and a 4-week rolling average — to predict next week's per-product demand, and validated it against a naive baseline using MAE on a holdout set.
- Wrote a data-ingestion layer that auto-detects which of two real vendor export formats was uploaded (transactional vs. aggregate snapshot), with a fuzzy column mapper and keyword-based category rules so the app degrades gracefully to a restock-priority ranking when no dated sales history is available.
- Added a rule-based Q&A assistant that answers questions about top sellers, revenue, and pricing directly from the loaded data, with no external LLM call or GPU needed.
- Deployed the app on Streamlit Community Cloud and used caching (st.cache_data/st.cache_resource) so switching between pages doesn't retrain the model each time.

Smart Travel Itinerary Planner (LLM, RAG, Gemini API) — github.com/SAILESHKUMAR33/Smart-Travel-Itinerary-Planner

- Built a Streamlit app that turns a city, a budget, and a list of interests into a day-by-day, 3-day travel itinerary using a retrieval-augmented generation pipeline on top of Google's Gemini API.
- Prompted Gemini to generate a pool of real, geographically-grounded points of interest as schema-constrained JSON, then used Gemini's embedding model plus cosine similarity to retrieve a balanced set per interest category, so results stay diverse instead of skewing toward one theme.
- Applied geospatial logic — haversine distance for clustering nearby POIs and a nearest-neighbor route builder — to keep each day's stops within one area and visited in a sensible order, and filtered candidates against a per-day budget.
- Added a local-specialties page that flags a city's single most iconic food or drink and a walkthrough page explaining the pipeline for demos, then deployed it on Streamlit Community Cloud with secrets-based API key handling.

EDUCATION

B.Tech, Artificial Intelligence and Data Science | Saveetha Engineering College, Chennai | CGPA: 7.0

2022 – 2026

SSLC | S.T. Britto Matric School | 79.8%

HSC | CEOA, Matric and Higher Sec School | 72%

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript

Machine Learning / Deep Learning: Machine Learning, Deep Learning, CNN, VGG16/VGG19, MLP, YOLOv8, RandomForest, scikit-learn, Model Training & Evaluation, Time-Series Forecasting

AI / NLP / Generative AI: NLP, Computer Vision (OpenCV, DeepSORT), Large Language Models (LLMs), Gemini API, Prompt Engineering, Embeddings, Semantic Search, Retrieval-Augmented Generation (RAG)

Tools & Platforms: Git, GitHub, VS Code, Streamlit, Power BI, React Native, Expo, Star UML

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Data Preprocessing, Feature Engineering, Model Evaluation (Accuracy, Precision, Recall, F1-score, ROC/AUC, MAE)

CERTIFICATIONS

- AI with Python — Infosys
- Python for Data Science — IBM, Great Learning
- Deep Learning, GPU Programming — Coursera

CORE COMPETENCIES

Communication • Teamwork • Leadership • Time Management • Analytical Thinking • Problem Solving

LANGUAGES

Tamil | English